EXPERIMENT – 10

PIR SENSOR IN PROTEUS USING ARDUINO

AIM: To write an Embedded C program for interfacing PIR Sensor using Arduino Uno and Proteus.

OBJECTIVES:

1. To interface the PIR sensor with Arduino Uno.
2. To detect motion using the PIR sensor output.
3. To indicate motion detection using an LED and serial message.

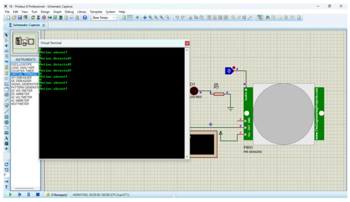
PROGRAM:

```
int Status = 13;
int sensor = 12;
void setup()
{
   Serial.begin(9600);
   pinMode(sensor, INPUT);
   pinMode(Status, OUTPUT);
}
void loop()
{
   long state = digitalRead(sensor);
   Serial.println(state);
   if(state == HIGH)
   {
      digitalWrite(Status, HIGH);
      Serial.println("Motion detected!");
      delay(1000);
   }
   else
   {
      digitalWrite(Status, LOW);
      Serial.println("Motion absent!");
      delay(1000);
   }
}
```

OUTPUT IMAGE:

```
void loop() {
 long state = digitalRead(sensor);
 Serial.println(state);
 if(state == HIGH) {
  digitalWrite (Status, HIGH);
  Serial.println("Motion detected!");
  delay(1000);
 }
 else {
  digitalWrite (Status, LOW);
  Serial.println("Motion absent!");
  delay(1000);
  }
}
```

OUTPUT:



RESULT:

Thus the program has been successfully verified and executed.

RESULT:

Thus the program has been successfully verified and executed.